

Order Hemiptera

Bugs

Bugs vary in size from species a few millimeters long to giant aquatic predators large enough to catch fishes and frogs. They are found all over the world, in every terrestrial habitat as well as in freshwater. The order comprises 134 families and 88,000 species. There are 4 suborders: Coleorrhyncha, consisting of a single family; Heteroptera or true bugs (including plant bugs); Auchenorrhyncha (including treehoppers, lantern bugs, and cicadas); and Sternorrhyncha (including whiteflies and scale insects). All bugs have 2 pairs of wings; in the Heteroptera, the front ones are typically larger and thickened. The mouthparts, which form a long rostrum, or “beak,” pierce and suck, and many sap-sucking species are serious crop pests. Metamorphosis is incomplete.

Family Aleyrodidae

Whiteflies

Occurrence 1,200 spp. worldwide, especially in warmer regions; on a wide range of plants, including crops



These delicate, mothlike bugs are 1/32–1/8 in (1–3 mm) long. They carry a pair of 7-segmented antennae, and 2 pairs of transparent, white, or mottled wings of similar size. Females lay their eggs on tiny stalks on the undersides of leaves. Many of these plant sap-suckers are serious pests.



TRIALEURODES VAPORARIORUM
The greenhouse whitefly is a common pest of cucumbers and tomatoes grown under glass. It may also attack field crops in warm conditions.

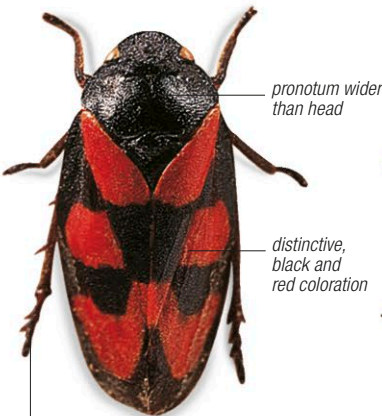
Family Cercopidae

Froghoppers

Occurrence 2,400 spp. worldwide, especially in warmer regions; on shrubs, trees, herbaceous plants



These squat, round-eyed, plant bugs are very good jumpers. Many species have drab coloration but some are colored in bright red and black or yellow and black patterns. Body length ranges from 3/16–3/4 in (0.5 to 2 cm). Eggs, up to 30 in some species, are laid in soil or in plant tissue. Nymphs feed on plant sap. They surround themselves in foam, which hides them and stops them from drying.



CERCOPIS VULNERATA
This froghopper inhabits grassy places and meadows in Europe and Asia.

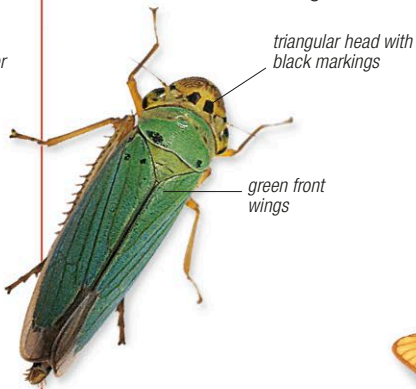
Family Cicadellidae

Leafhoppers

Occurrence 16,000 spp. worldwide; almost anywhere with vegetation, abundant in agricultural habitats



Leafhoppers are typically slender-bodied and 1/8–3/4 in (0.3–2 cm) long. Many species are brown or green; some may be brightly striped or spotted. Males attract females with low-amplitude courtship calls, which travel through leaves and stems, instead of being carried through the air. Once a female has mated, she lays hundreds of eggs over a period of 6–8 weeks. Leafhoppers are highly destructive pests, attacking many of the world's staple crops, such as rice, corn, cotton, and sugarcane.



CICADELLA VIRIDIS
Widespread in Europe and Asia, the common green leafhopper can be a pest of fruit, rice, wheat, and sugarcane.

Family Aphididae

Common aphids

Occurrence 2,250 spp. worldwide, especially in northern temperate regions; on most wild and cultivated plants



Their huge reproductive potential makes aphids among the most destructive of all plant-eating insects, affecting almost all crop species by feeding and transmitting diseases. Also called greenflies or blackflies, these are pear-shaped, soft-bodied insects, mostly 3/16 in (5 mm) long, and typically green, pink, black, or brown. The abdomen usually has a pair of cornicles, short

tubes that emit a defensive secretion. Reproduction is often parthenogenetic or only seasonally sexual.



MACROSIPHUM ALBIFRONS
The American lupin aphid is a pest of garden flowers, both in North America and Europe. Here, one female (top left) is giving birth.

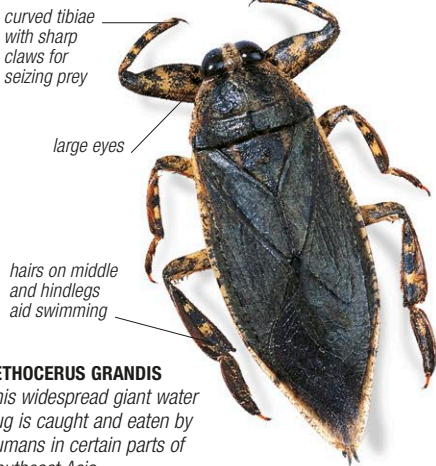
Family Belostomatidae

Giant water bugs

Occurrence 150 spp. worldwide, especially in tropical and subtropical regions; in standing water, slow-moving streams



Also known as electric light bugs due to their attraction to lights, this family contains the largest bugs, able to eat frogs and fishes. They have a brownish, oval body that is 1/2–4 in (1.5–10 cm) long. The front legs are modified as prey-capturing pincers. Uniquely among insects, females glue their eggs to the male's back, who then broods them.



LETHOCERUS GRANDIS
This widespread giant water bug is caught and eaten by humans in certain parts of Southeast Asia.

Family Cicadidae

Cicadas

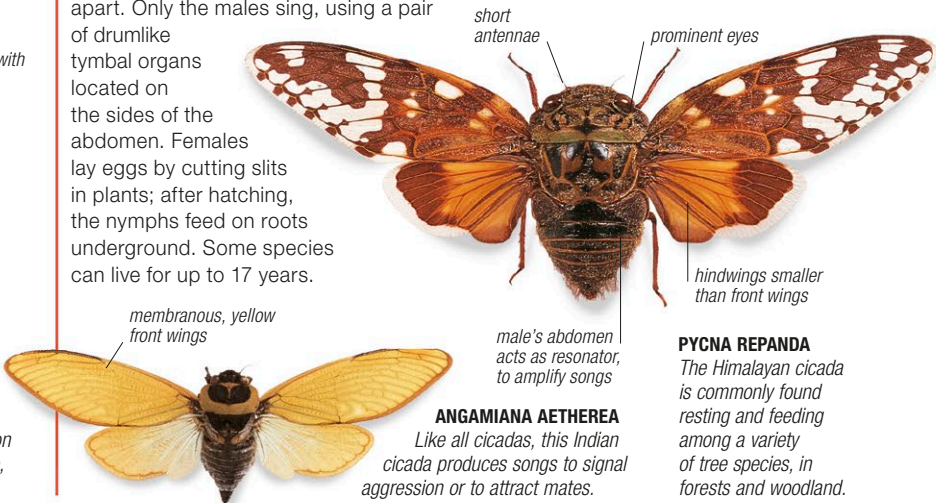
Occurrence 2,500 spp. worldwide in warmer regions; on suitable host trees or shrubs



Cicadas include some of the world's noisiest insects, producing calls that can be heard up to 1 mile (1.5 km) away. The adults are usually dark brown or green, with squat bodies 3/4–2 1/4 in (2.2–5.5 cm) long. They have short antennae, and prominent eyes set far apart. Only the males sing, using a pair of drumlike tymbal organs located on the sides of the abdomen. Females lay eggs by cutting slits in plants; after hatching, the nymphs feed on roots underground. Some species can live for up to 17 years.



CICADA ORNI
This species, common in southern Europe and the Mediterranean, has a grayish bloom, clear front wings with 11 black spots, and a long rostrum. The adults are eaten by birds such as jays.



ANGAMIANA AETHEREA
Like all cicadas, this Indian cicada produces songs to signal aggression or to attract mates.

PYCNA REPANDA
The Himalayan cicada is commonly found resting and feeding among a variety of tree species, in forests and woodland.